Homework — 1.1.2 Types of Processor — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] — 1 mark per difference (max 3): RISC has a **small set of simple, fixed-length** instructions vs CISC's **large set of complex, variable-length** instructions; RISC instructions typically complete in **one clock cycle** vs CISC instructions taking **several cycles**; RISC needs **more RAM** / longer programs (more instructions to do the same job) vs CISC's more compact code; RISC relies on the **compiler/software** to build complex operations, CISC does it in **hardware/circuitry**; RISC processors tend to use **less power** / produce less heat.

2. [2] — 1 mark per valid reason (max 2), linked to a phone: RISC uses **less power** → longer battery life; simpler instructions run efficiently/cool in a small device without a fan; easier to pipeline → good performance per watt; cheaper/simpler hardware keeps the handset cost down. *(Generic "RISC is simpler" with no link caps at 1.)*

3. [2] — 1 mark: RISC has **few, simple, fixed-length** instructions (most aiming for one cycle). 1 mark: this makes RISC **easier to pipeline** (predictable timing/length), whereas CISC's complex, variable-length, multi-cycle instructions are **harder to pipeline**.

4. [3] — 1 mark: a GPU has a **very large number of (simpler) cores** (far more than a CPU). 1 mark: each core carries out the **same instruction on a different item of data** at the same time. 1 mark: so a calculation that would be done **sequentially** on a CPU is done **in parallel** across the data, giving much higher throughput (e.g. transforming every pixel/vertex at once).

5. [2] — 1 mark per reason (max 2): parts of the task **depend on the results of other parts** (data dependencies) so must run sequentially; **coordination/communication overhead** of splitting the work between cores and combining results; some of the program **cannot be parallelised** and must run on one core; cores **compete for shared resources** (memory/cache/bus).

6. [2] — 1 mark each (max 2): training machine-learning / neural-network models; scientific modelling and simulation (e.g. weather, protein folding); data mining / analysing large data sets; breaking encryption or mining cryptocurrency (repeated hashing); audio/video encoding.

Part B (6 marks)

7. [3] — 1 mark: Von Neumann uses a **single shared memory and bus** for both data and instructions. 1 mark: so it **cannot fetch an instruction and read/write data at the same time** (they compete for the one bus) — the bottleneck. 1 mark: Harvard uses **separate memories and buses** for data and instructions, so instruction fetch and data access can happen **simultaneously**.

8. [2] — 1 mark each (factor + effect), max 2: **clock speed** — more pulses/sec → more FDE cycles/sec; **cache size** — more frequently used data/instructions held near the CPU → fewer slow RAM fetches; **pipelining** — overlaps FDE stages → higher throughput. *(Accept word/bus width.)*

9. [1] — **CIR** (Current Instruction Register).

Total: 14 + 6 = 20